

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester V Examination, 2013-14

Paper No. GLB-502: Igneous Petrology, Mineralogy and Crystallography

Time: 3 Hours

Maximum Marks: 70

Ref. No.: BS/Sem V/258

*Instructions: Attempt any **five** questions, selecting at least **two** from each of Sections A and B. Section C is compulsory. All questions carry equal marks (14 marks each).*

SECTION — A

Question 1. Give the salient features of mineralogical classification schemes for plutonic igneous rocks. [14]

Question 2. Write the petrographic description of **any four** of the following: $[4 \times 3.5 = 14]$

- (a) Alkali Granite
- (b) Peridotite
- (c) Komatiite
- (d) Dacite
- (e) Norite
- (f) Rhyolite

Question 3. Write descriptive notes on the following: $[2 \times 7 = 14]$

- (a) Total Alkali-Silica (TAS) classification
- (b) Physical properties affecting the nature of magma

Question 4. Describe with the help of a neat sketch the crystallization of magma in the phase equilibrium of Diopside-Albite-Anorthite system. Comment on its petrogenetic significance. [14]

SECTION — B

Question 5. Describe the symmetry and forms belonging to the holosymmetric class of the Triclinic system and draw a suitable stereogram. Name any four minerals belonging to this class. [14]

Question 6. Write short notes on **any two** of the following: $[2 \times 7 = 14]$

- (i) Classification of minerals

(ii) Isomorphism and Polymorphism

(iii) Pyroxene group of minerals

Question 7. Write a detailed note on the structural classification of silicates. [14]

SECTION — C (Compulsory)

Question 8. Answer the following questions (in two or three sentences, or by filling in the blanks as indicated): [7 × 2 = 14]

(i) What is secondary magma?

(ii) What is a rock series?

(iii) What are the petrographic differences between syenite and alkali feldspar syenite?

(iv) The most abundant element in the earth's crust is _____ whereas the most abundant mineral in the Earth's crust is _____.

(v) What is relief in optical mineralogy?

(vi) Name two minerals showing pleochroism.

(vii) Name two biaxial minerals.